

SATQUERY AI - P4 REMAINING WORK

Kavin | Geospatial / Data Engineer | Assessment date: 13 September 2026 | SIH deadline: 20 September 2026

Scope: Bhoonidhi/STAC, normalization, COG, GDAL/Rasterio, PostGIS, TITiler

Assessment basis: Repomix repository snapshot supplied in chat + the six role PRDs + Master Integration PRD + repository audit reports dated 13 Sep 2026. Live GitHub branch state could not be fetched in this environment, so this is a snapshot-based assessment.

~75-80% local
implementation

	12	8	0	24-32h
Issue-level readiness signal	Done	Partial	Not ready	Estimated remaining

1. Executive verdict

P4 has substantial real geospatial/provider code already. The key remaining work is integration hygiene: unify duplicate search paths, prove live provider retrieval, finish map tile serving and robust production failure semantics, then hand P3/P5 clean live artifacts.

2. What is actually present in the repository

- **Real provider code exists:** Bhoonidhi HTTP auth/search/download adapter and Pystac STAC adapter are present.
- **Geo stack exists:** raster validation, reprojection, AOI clipping, COG, PostGIS helpers, Redis/Celery worker and TITiler integration path are present.
- **Integration debt:** there are duplicate SearchService implementations; live verification and the final P3/P5 artifact path remain unresolved.

3. Issue-by-issue remaining work

P4-01 EO/Data service skeleton and provider adapter interface	DONE	EO service, provider base and adapters exist.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-02 Provider configuration and secret boundary	DONE	Provider config/secrets boundaries exist.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-03 Observation and AssetRef domain model	DONE	Canonical observation/scene models and validation exist.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-04 STAC adapter	DONE	Pystac-client based STAC adapter exists; live verification remains.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-05 Bhoonidhi adapter	DONE	Real HTTP auth/search/download adapter exists; live runtime/stress verification remains.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-06 Observation normalization pipeline	DONE	Normalization logic is implemented.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-07 Spatial/temporal observation search	PARTIAL	Real-ish implementation exists, but there are duplicate SearchService implementations and integration drift.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-08 Asset resolver	DONE	Resolver exists; it needs full agent/inference integration verification.	No substantive implementation work expected; verify with integrated test/acceptance.

P4-09 Secure asset retrieval and content validation	PARTIAL	URL/raster validation exists; end-to-end secure retrieval needs stronger tests.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-10 Raster validation	DONE	Raster validation boundaries and tests exist.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-11 CRS normalization and reprojection	DONE	Real CRS/reprojection utilities exist.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-12 AOI clipping and windowed processing	DONE	Implementation exists; user-defined AOI integration needs end-to-end validation.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-13 COG generation and overviews	DONE	COG utilities/processing path exists.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-14 PostGIS spatial operations	PARTIAL	Spatial utilities exist; production persistence/use in the flagship path is incomplete.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-15 TiTiler integration	PARTIAL	Tile integration is conditional/incomplete; real map-serving path needs hardening.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-16 Async GeoJob worker	DONE	Redis/Celery-oriented async worker exists.	No substantive implementation work expected; verify with integrated test/acceptance.
P4-17 Geo failure recovery and fixture fallback	PARTIAL	Fallback exists, but final behavior must report degraded state rather than fabricate success.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-18 Monitoring observation selection support	PARTIAL	Monitoring hooks exist but need integrated scheduled-run selection.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-19 EO data quality scoring/enrichment	PARTIAL	Quality scoring exists; must feed sensor arbitration and evidence.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.
P4-20 Pinned EO/Geo fixture pack and release hardening	PARTIAL	Fixtures exist, but live integration and release verification remain.	Finish the production integration, hardening, tests and acceptance evidence described by the issue.

4. Critical execution order (13-20 Sep)

- 13 Sep: unify SearchService path and align endpoint schema with P2/P3/P5.
- 14 Sep: prove live Bhoonidhi/STAC search + asset resolution with real credentials/config.
- 15 Sep: validate secure retrieval, AOI clip, COG and PostGIS against realistic rasters.
- 16 Sep: finish real TiTiler/tiles path and GeoJSON/mask handoff.
- 17 Sep: connect monitoring selection and quality scoring to P2.
- 18 Sep: release/pinned data profile; remove accidental fixture dependence from production path.
- 19-20 Sep: integration/reliability only.

5. Dependencies and hand-offs

Primary cross-team contacts: P1 for API/error/auth contract; P2 for search/asset selection contract; P3 for GeoJob/asset requirements; P5 for tiles/GeoJSON.

6. Definition of Done for this role

- Production code path works without fabricated business results.
- Unit tests cover core logic; contract tests cover all exposed schemas; integration tests exercise actual service boundaries.
- Security controls are enforced by the real request path, not merely by test fixtures.

- Trace IDs / mission IDs / run IDs / organization IDs propagate correctly where applicable.
- Failures are explicit and recoverable. A failed external dependency must not be converted into a fake successful result.
- No hard-coded real-world analysis result is used to satisfy the flagship path.
- Acceptance evidence is attached to the PR and the issue is only closed after integrated verification.

7. Critical rule: fixtures, mocks and fallback behavior

Mocks/fixtures are test infrastructure, not product implementation. They are allowed for unit tests, contract tests, CI isolation and temporary developer unblock. They must not replace required production integrations or fabricated analysis results. When a live dependency is unavailable, the final application should surface an explicit degraded/abstained state with provenance rather than silently claiming success.

8. Snapshot limitations and confidence

The supplied Repomix is a read-only snapshot; binary files and gitignored paths are not included. The repository itself documents that Repomix can exclude files via .gitignore. The current snapshot also contains audit reports whose P5 assessment conflicts with the actual visible app/components in the same snapshot; this report therefore uses the code structure and targeted code inspection over the stale P5 audit conclusion. GitHub live branch state was not independently verified in this environment.